

ODD END SEMESTER EXAMINATION DECEMBER-2025

Roll no.....

Name of the Course: B. Tech (CE) / BSc (H) Geology

Semester: III

Name of the Paper: Geomatics Engineering / Basic Surveying Paper Code: TCE 302 / BSGT 311

Time: 3 Hour

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub question among a, b & C in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1.	(20 Marks), CO1
a. Differentiate between the prismatic compass and the surveyor's compass in terms of instrument construction, method of taking observations, and field applications	
b. Determine the whole circle bearing of a survey line when the <i>latitude and departure</i> are given as +90 m and -35.1 m respectively	
c. Explain, with suitable illustrations, the functions of main survey stations, base line, and check or tie lines in a chain survey, emphasizing their importance for maintaining survey accuracy.	
Q2.	(20 Marks), CO2
a. Describe the trilateration method of surveying, and evaluate its <i>advantages and limitations</i> compared to other control establishment methods.	
b. A 20 m chain was found to be 10 cm too long. If a line measures 250 m with this chain, compute the true length of the line.	
c. Illustrate the concept of triangulation with neat sketches and justify its use in forming horizontal control networks.	
Q3.	(20 Marks), CO3
a. Identify and discuss the various sources of error in leveling, and explain how such errors can be minimized through proper field practice.	
b. The following staff readings (in meter) were observed with a dumpy level: 5.21, 4.50, 7.24, 8.42, 9.51, 7.13, 6.91, 8.26, 9.71, 10.21. The instrument was shifted after the 3rd, 6th, and 9th readings. If the Reduced Level (RL) of the first station is 100 ft, prepare a level field book (using the <i>collimation method</i>) and apply the usual arithmetic check.	
c. Define contours and describe their characteristics using appropriate sketches to show how they represent terrain features.	
Q4.	(20 Marks), CO4
a. Explain the concept of the two-point problem in plane table surveying and outline the steps involved in determining the instrument position graphically.	
b. List and explain the common errors encountered in plane table surveying and propose suitable methods for their elimination.	
c. Define a Digital Elevation Model (DEM) and discuss different methods for its generation, highlighting the major geospatial applications of DEMs.	

Q5.

- Determine the most probable value of the angle θ .